

Ex.no:5	DATA VISUALIZATION USING MATPLOTLIB

Aim:

To visualize the features and distributions of the **Tourism Dataset** using line charts, bar charts, pie charts, histograms, scatter plots, box plots, and multiple line charts using Matplotlib.

ALGORITHM 1: LINE CHART

Objective: To visualize the number of tourism destinations across different continents.

Steps

1. Load the Tourism dataset.
2. Count the number of destinations in each continent.
3. Plot the destination counts using a line chart.
4. Add a title and axis labels.
5. Display the graph.

Interpretation

The line chart shows the distribution of tourism destinations across different continents and helps compare tourism availability among continents.

ALGORITHM 2: BAR CHART

Objective: To compare the number of destinations for different tourism types.

Steps

1. Select the Tourism_Type column.
2. Count the number of destinations for each tourism type.
3. Create a bar chart using tourism types and their counts.
4. Add a title and axis labels.
5. Display the chart.

Interpretation

The bar chart provides a comparison of the number of tourism destinations available under different tourism types.

ALGORITHM 3: PIE CHART

Objective: To visualize the distribution of accommodation types.

Steps

- Select the Accommodation_Type column.
- Count the occurrences of each accommodation type.
- Calculate their proportions.
- Create a pie chart.
- Display percentage labels.
- Add a suitable title.

Interpretation

The pie chart shows the percentage distribution of different accommodation types available in the Tourism Dataset.

ALGORITHM 4: HISTOGRAM

Objective:

To visualize the distribution of estimated tourism budgets.

Steps

- Select the Estimated_Budget_USD column.
- Divide the budget values into intervals or bins.
- Create a histogram using Matplotlib.
- Add a title and axis labels.
- Display the histogram.

Interpretation

The histogram shows the frequency distribution of estimated tourism budgets and helps understand the budget range of different destinations.

ALGORITHM 5: SCATTER PLOT

Objective: To analyze the relationship between average duration of stay and estimated budget .

Steps

- Select Average_Duration_Days as the X-axis variable.
- Select Estimated_Budget_USD as the Y-axis variable.
- Plot the observations using a scatter plot.
- Add a title and axis labels.
- Display the graph.

Interpretation

The scatter plot helps identify the relationship between the average duration of a trip and the estimated tourism budget.

ALGORITHM 6: BOX PLOT

Objective: To visualize the distribution and identify possible outliers in the estimated tourism budget.

Steps

- Select the `Estimated_Budget_USD` column.
- Remove missing values if present.
- Create a box plot using Matplotlib.
- Add a suitable title and axis label.
- Display the plot.

Interpretation

The box plot shows the central distribution of estimated tourism budgets and helps identify unusually high or low budget values .

ALGORITHM 7: MULTIPLE LINE CHART

Objective: To compare the average estimated budget and average rating across different tourism types.

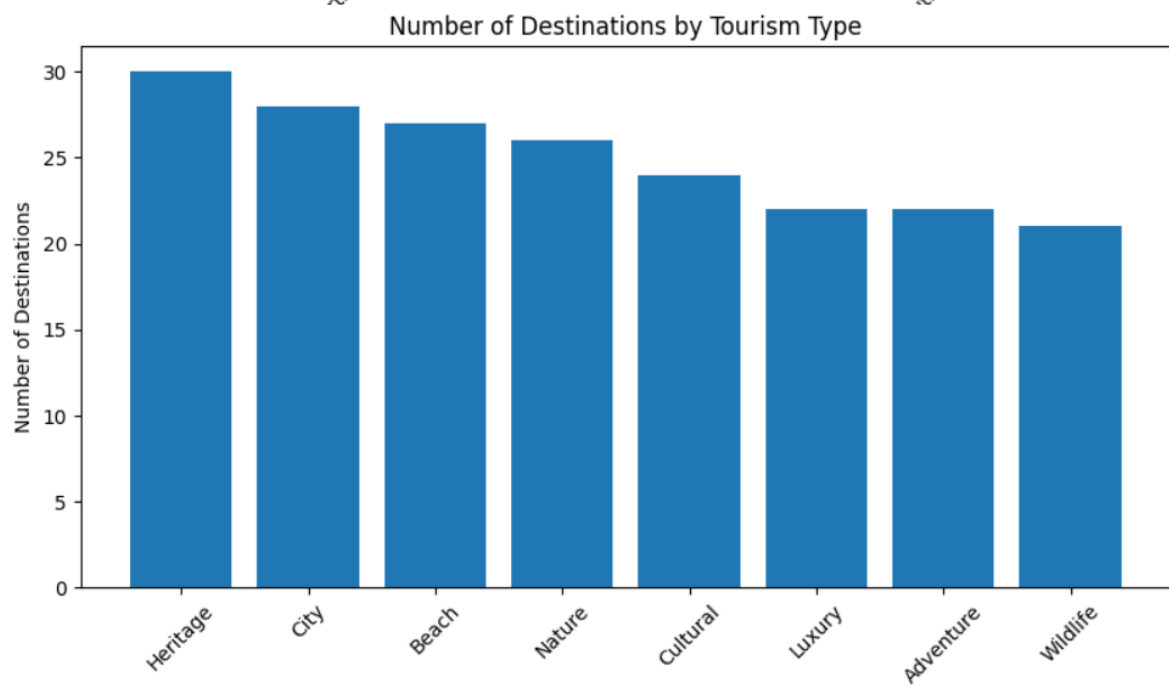
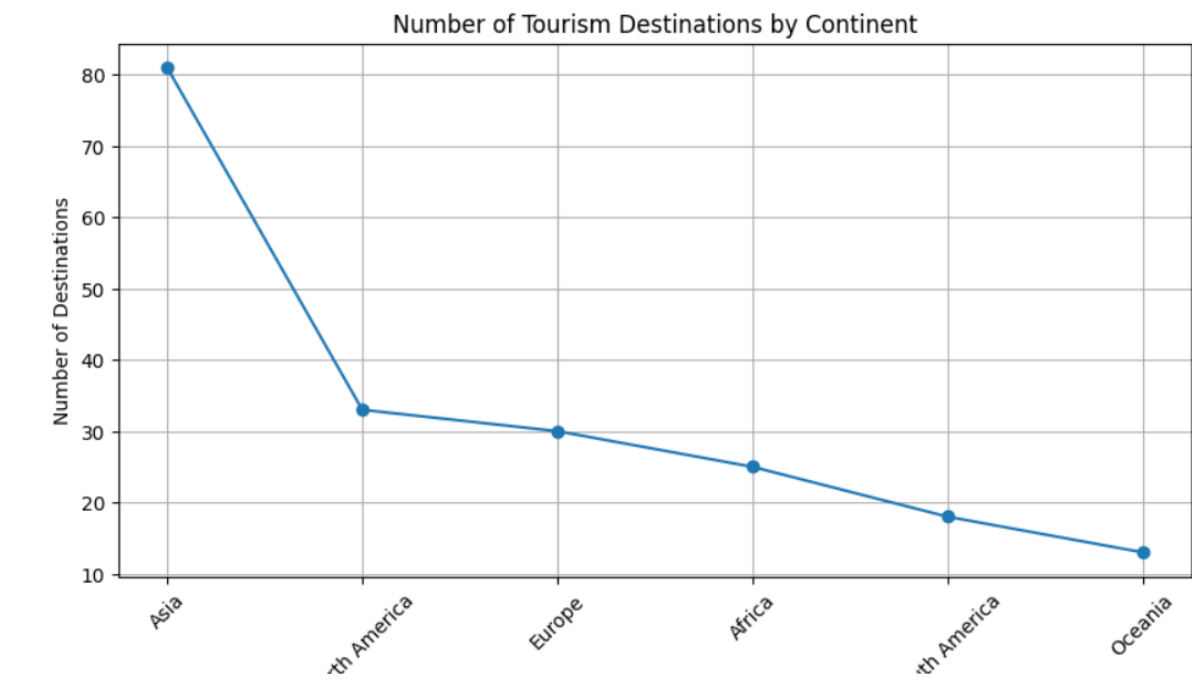
Steps

- Group the dataset based on tourism type.
- Calculate the average estimated budget for each tourism type.
- Calculate the average rating for each tourism type.
- Plot the average budget using a line chart.
- Plot the average rating using another line on the same graph.
- Add labels and a legend.
- Display the graph.

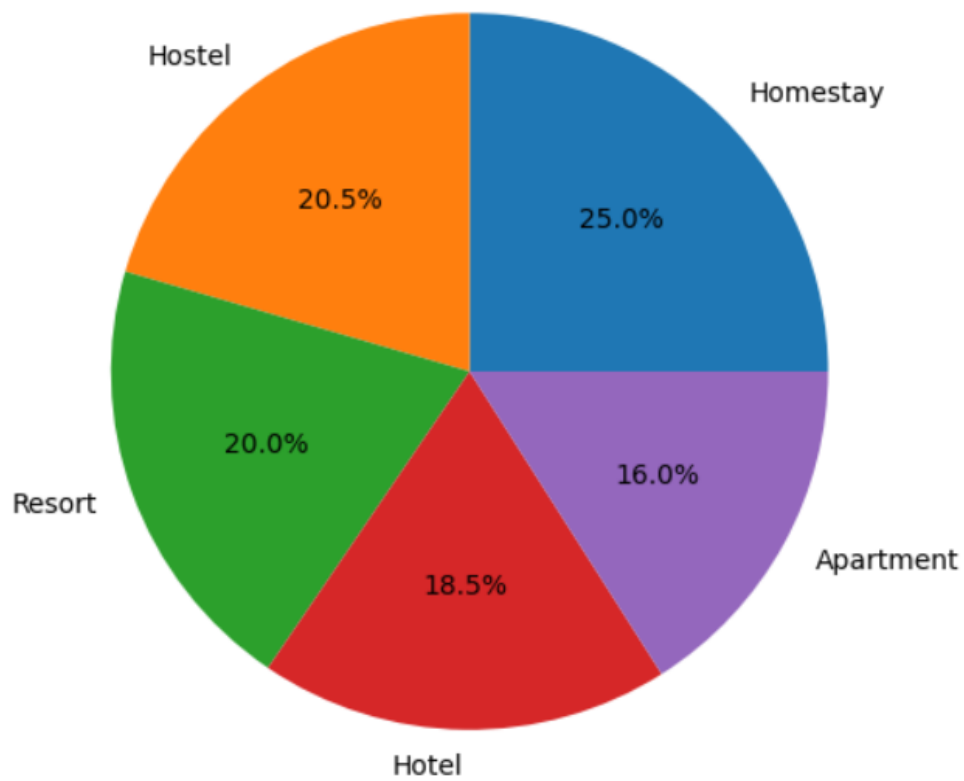
Interpretation

The multiple line chart allows comparison of different tourism characteristics across tourism types.

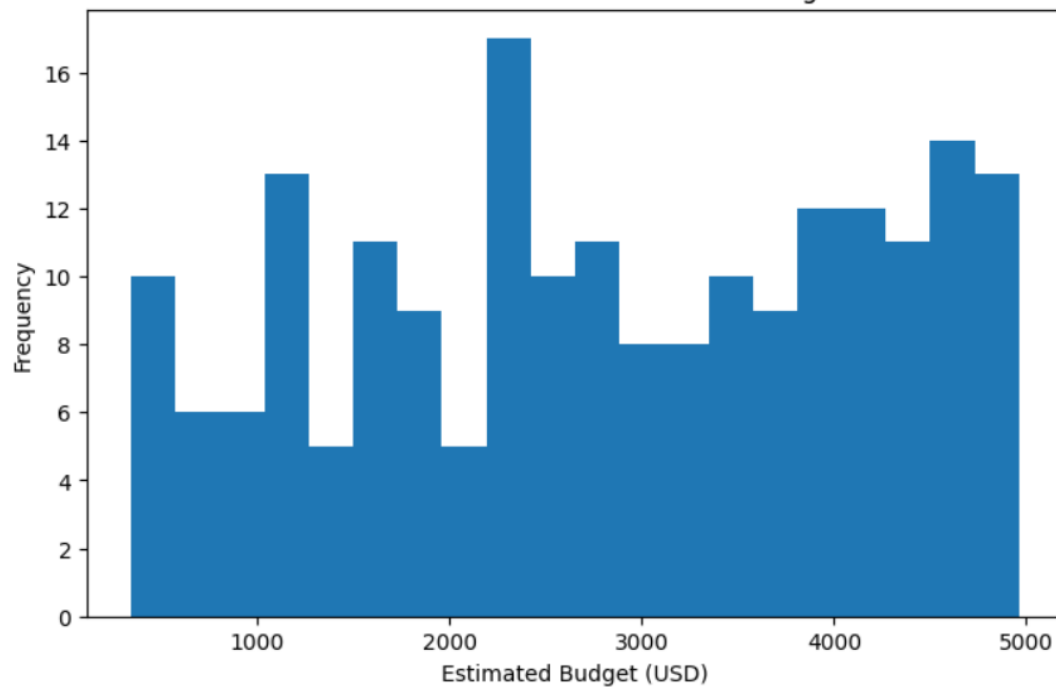
Output:

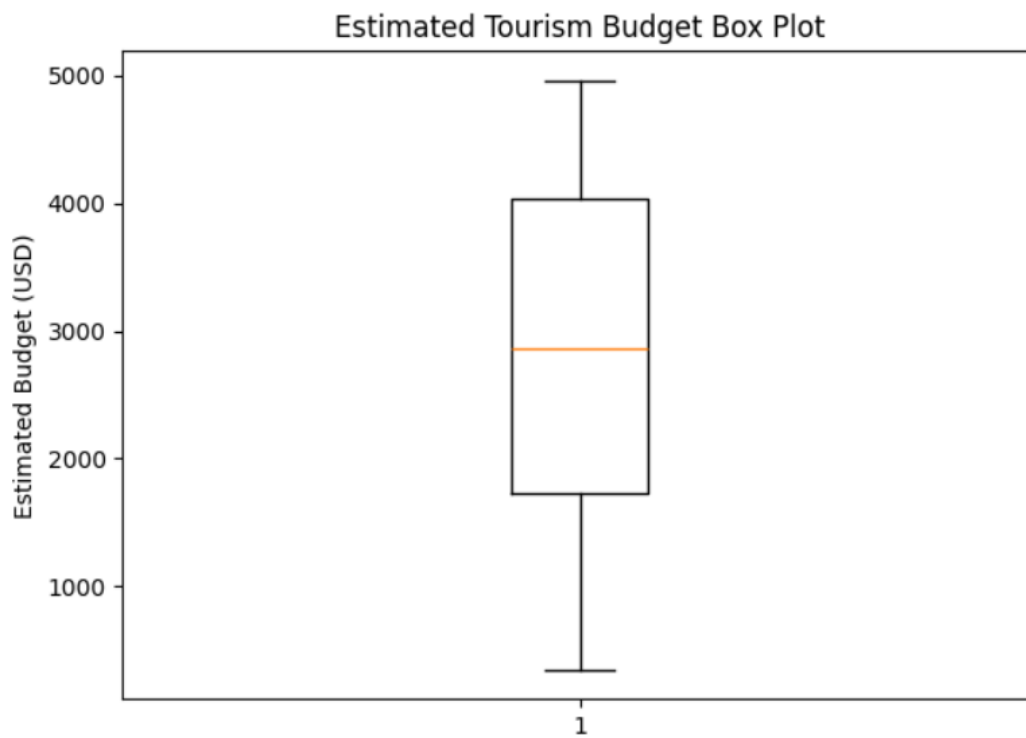
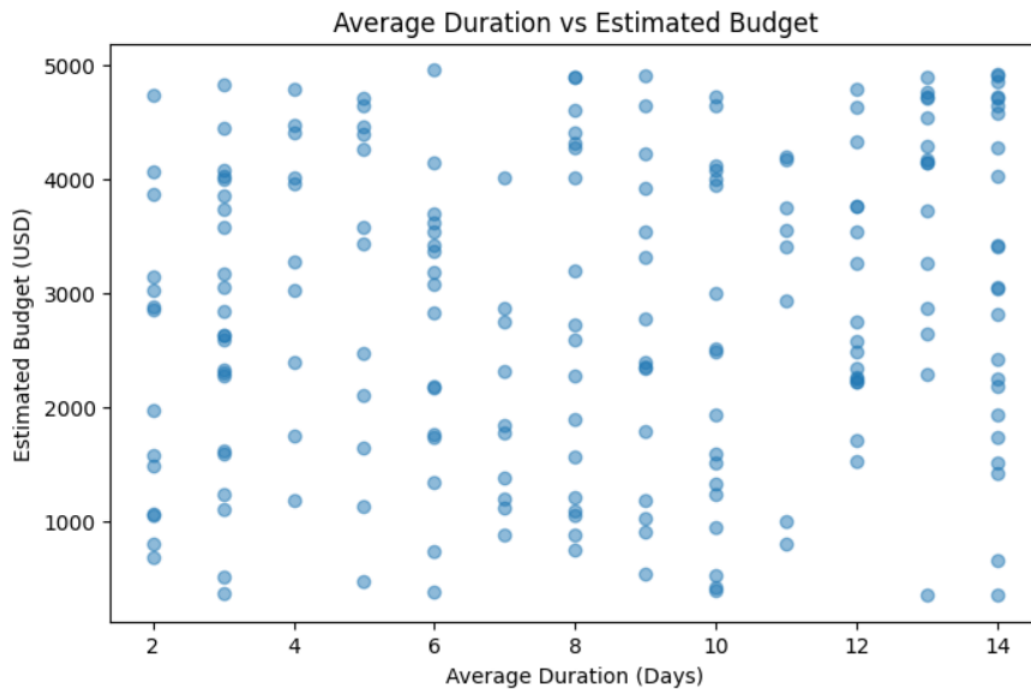


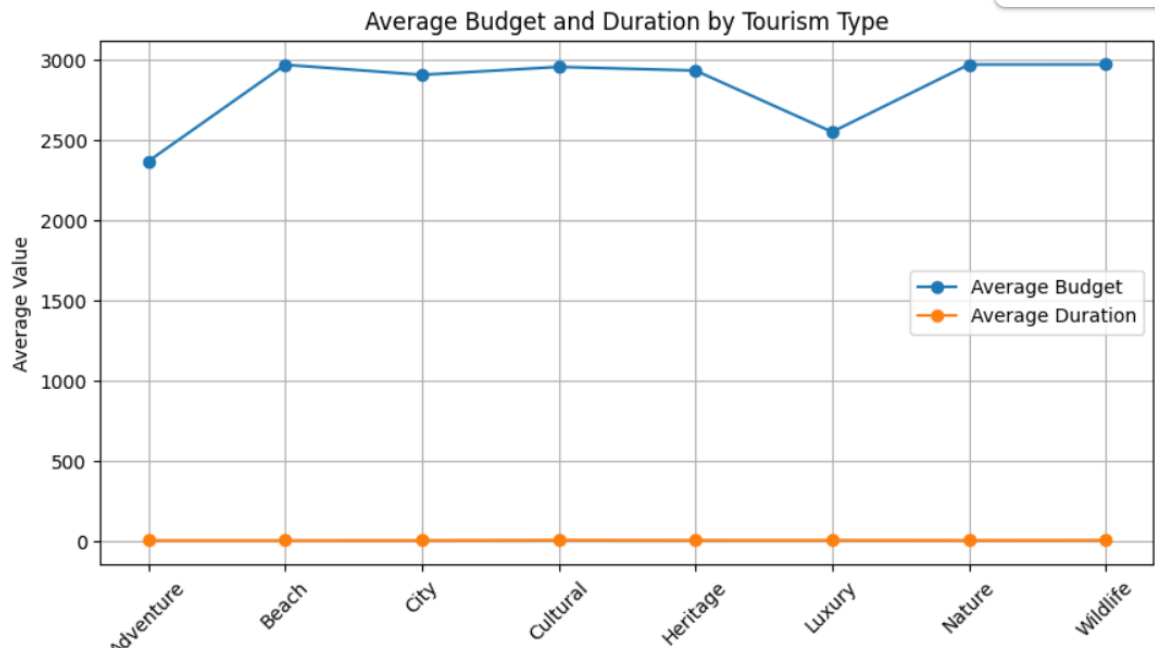
Accommodation Type Distribution



Distribution of Estimated Tourism Budget







Result:

Thus, the **Tourism Dataset** was successfully visualized using **Matplotlib**. Different visualization techniques such as **line charts, bar charts, pie charts, histograms, scatter plots, box plots, and multiple line charts** were implemented successfully to analyze the distribution and relationships among the tourism data attributes.